



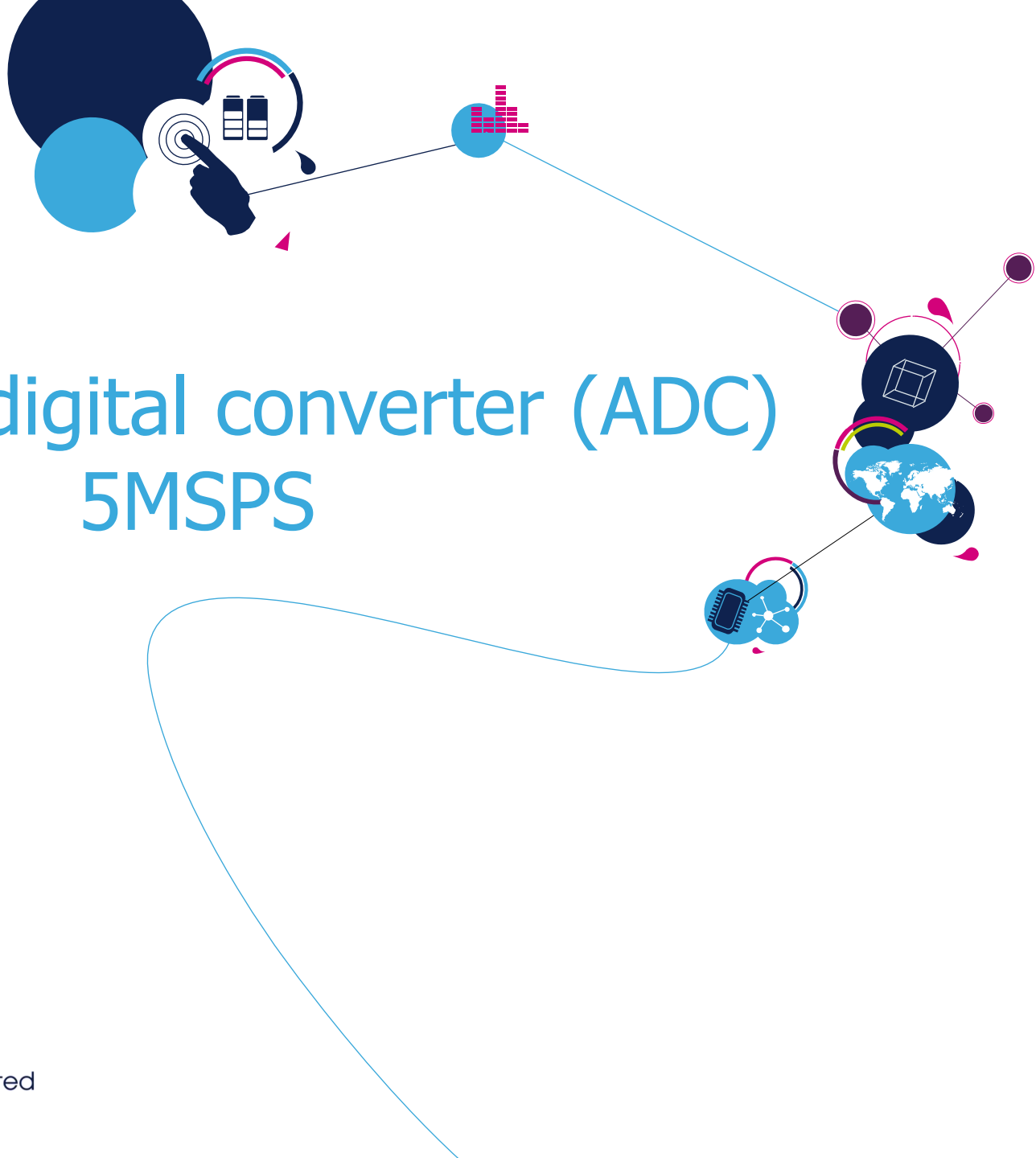
# STM32F3 Technical Training

**For reference only**

**Refer to the latest documents for details**



# STM32F30x Specific features/peripherals



# Analog-to-digital converter (ADC) 5MSPS

# ADC Features (1/2)

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- Up to 4 ADCs:
  - ADC1 & ADC2 are tightly coupled and can operate in dual mode (ADC1 is master)
  - ADC3 & ADC4 are tightly coupled and can operate in dual mode (ADC3 is master)
- Programmable Conversion resolution : 12, 10, 8 or 6 bit
- External Analog Input Channels for each of the 4 ADCs:
  - 5 fast channels from dedicated GPIOs pads
  - Up to 11 slow channels from dedicated GPIOs pads
- ADC conversion time:
  - Fast channels : up to 5.1Ms/s with 12 bit resolution in single mode
  - Slow channels: up to 4,8Ms/s with 12 bit resolution in single mode
- AHB Slave Bus interface
- Channel-wise programmable sampling time
- Self-calibration
- Configurable regular and injected channels
- Hardware assistant to prepare the context of the injected channels to allow fast context switching
- Can manage Single-ended or differential inputs

# ADC Features (2/2)

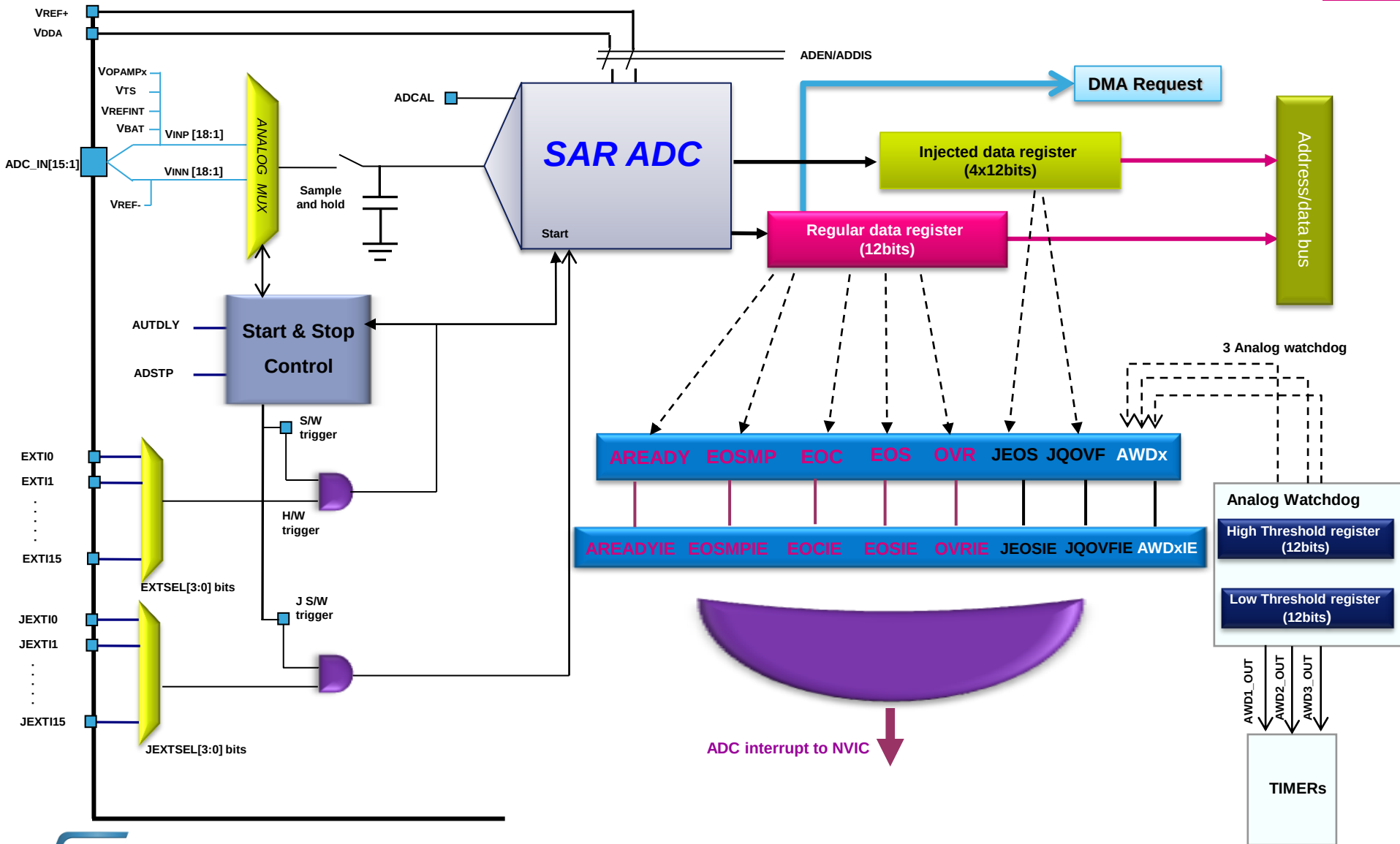
5

- 3 internal channels connected to :
  - Temperature sensor Vsense connected to ADC1
  - Internal voltage reference VREFINT connected to all ADCs
  - VBAT/2 power supply connected to ADC1
- Programmable sampling time
- Single, continuous and discontinuous conversion modes
- Dual ADC mode
- Left or right Data alignment with inbuilt data coherency
- Software or Hardware start of conversion
- 3 Analog Watchdog per ADC
- DMA capability
- Auto Delay insertion between conversions
- Interrupt generation

| Name        | Signal Type                                 | Remarks                                                                                                                                                     |
|-------------|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| VREF+       | Input, analog reference positive            | The higher/positive reference voltage for the ADC, $1.8\text{ V} \leq V_{\text{REF+}} \leq V_{\text{DDA}}$                                                  |
| VDDA        | Input, analog supply                        | Analog power supply equal to VDD and $1.8\text{ V} \leq V_{\text{DDA}} \leq V_{\text{DD}} (3.6\text{ V})$                                                   |
| VREF-       | Input, analog reference negative            | The lower/negative reference voltage for the ADC, $V_{\text{REF-}} = V_{\text{SSA}}$                                                                        |
| VSSA        | Input, analog supply ground                 | Ground for analog power supply equal to Vss                                                                                                                 |
| VINP[18:1]  | Positive input analog channels for each ADC | Connected either to external channels: ADC_INi or internal channels.                                                                                        |
| VINN[18:1]  | Negative input analog channels for each ADC | Connected to VREF- or external channels: ADC_INi-1                                                                                                          |
| ADCx_IN16:1 | External analog input signals               | Up to 16 analog input channels (x=ADC number = 1,2,3 or 4): <ul style="list-style-type: none"> <li>• 5 fast channels</li> <li>• 11 slow channels</li> </ul> |

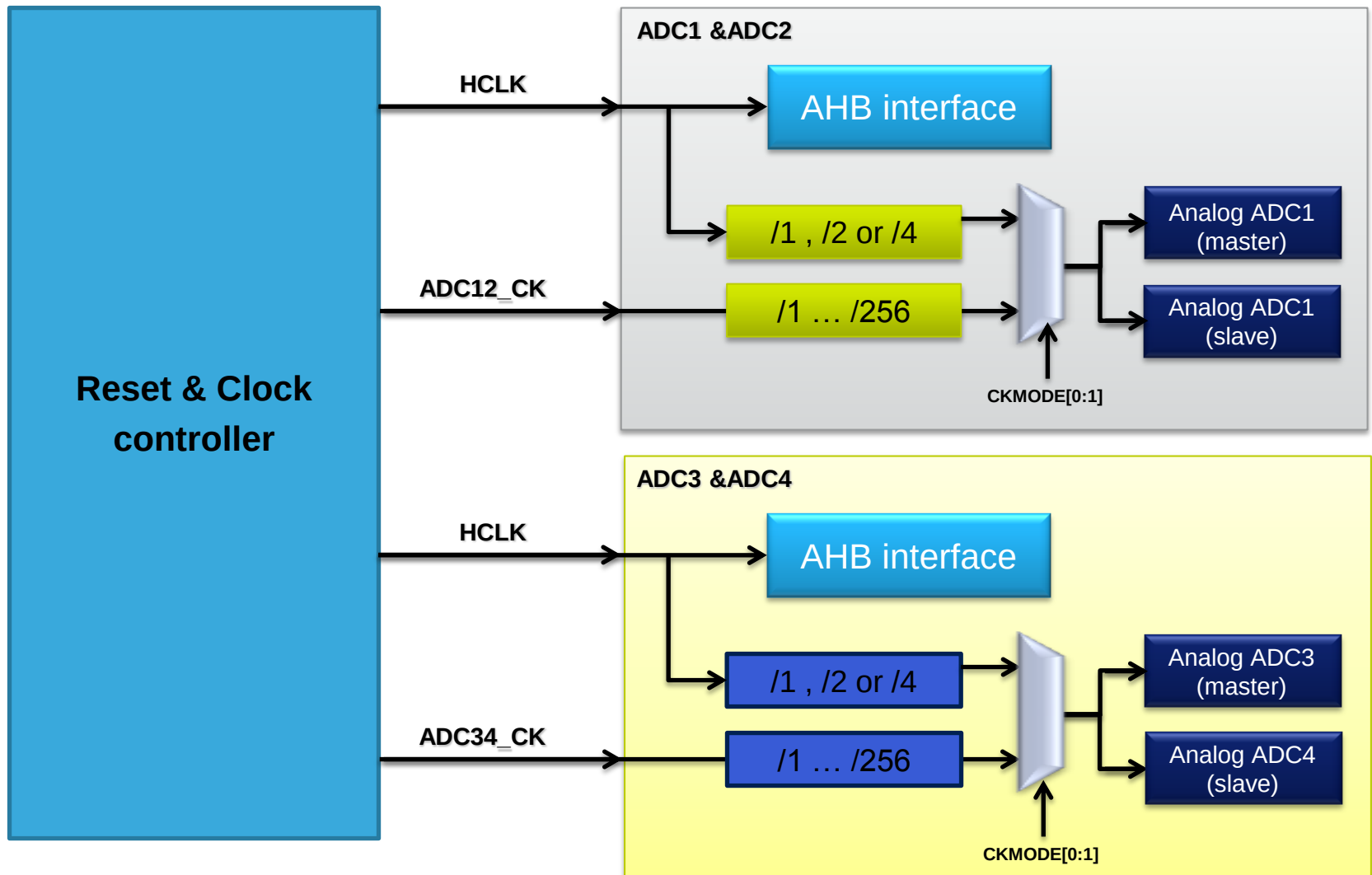
# ADC Block Diagram

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# ADC Clocks

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# How to choose ADC Clock

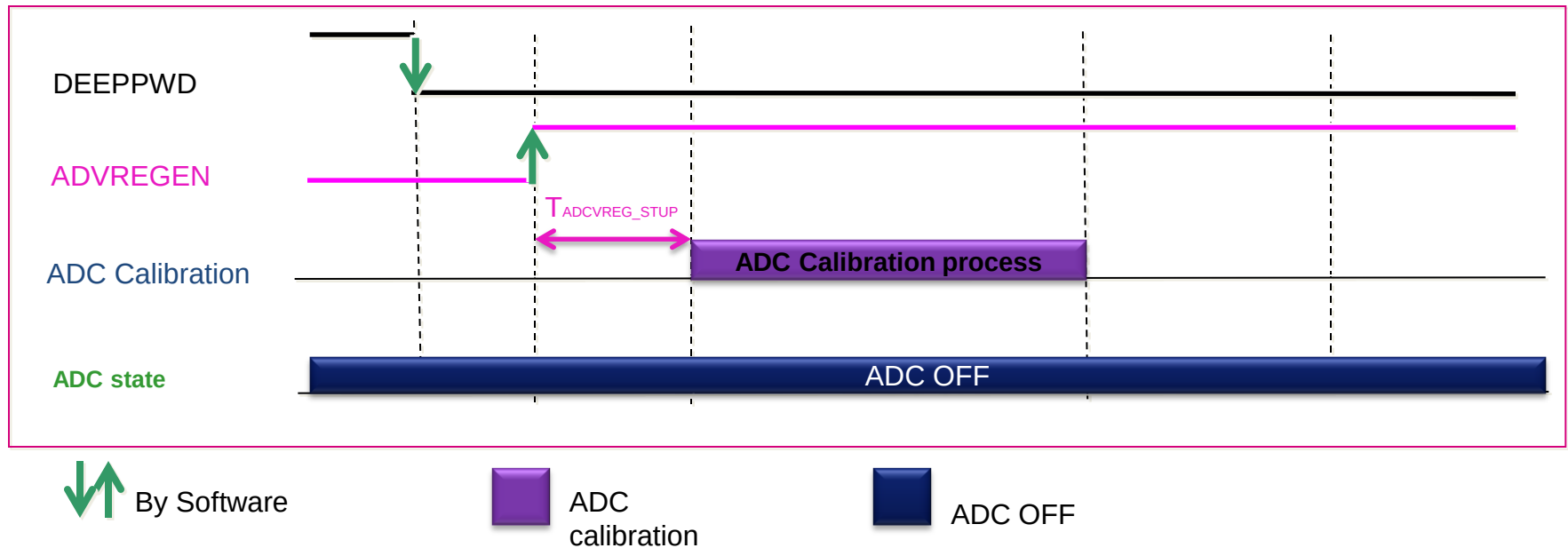
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| ADC clock source                               | ADCxy_CK                                                                                                                                                                                                                                                                                                                                        | AHB div 1, 2 or 4                                                                                                                                                          |
|------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Benefits                                       | <ul style="list-style-type: none"><li>Independent and asynchronous ADC clock versus AHB clock</li></ul>                                                                                                                                                                                                                                         | <ul style="list-style-type: none"><li>Bypassing the clock domain resynchronizations: deterministic latency between the trigger event and the start of conversion</li></ul> |
| Drawbacks                                      | <ul style="list-style-type: none"><li>Uncertainty of the trig instant is added by the resynchronizations between the two clock domains</li></ul>                                                                                                                                                                                                | <ul style="list-style-type: none"><li>ADC clock depends on the AHB clock</li></ul>                                                                                         |
| Clock constraints when using injected channels | <ul style="list-style-type: none"><li><math>F_{HCLK} \geq F_{ADC} / 4</math> if the resolution of all channels are 12-bit or 10-bit</li><li><math>F_{HCLK} \geq F_{ADC} / 3</math> if there are some channels with 8 bits resolution</li><li><math>F_{HCLK} \geq F_{ADC} / 2</math> if there are some channels with 6 bits resolution</li></ul> |                                                                                                                                                                            |

# ADC Deep-Power-Down Mode

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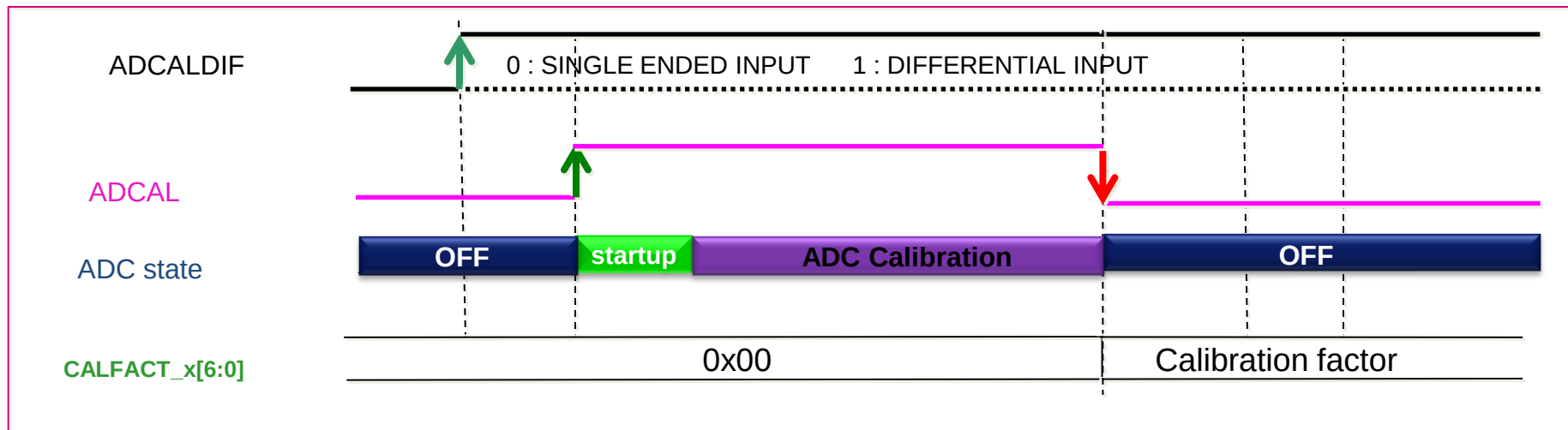
- By default, the ADC is placed in deep-power-down mode where its supply is internally switched off to reduce the leakage currents,
- To start ADC operations the following sequence should be applied:



# ADC Calibration

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- The calibration factor to be applied for single-ended input conversions is different from the factor to be applied for differential input conversions:
  - If `ADCALDIF=0`, calibration applied for single conversion and value stored in `CALFACT_S`
  - If `ADCALDIF=1`, calibration applied for differential conversion and value stored in `CALFACT_D`

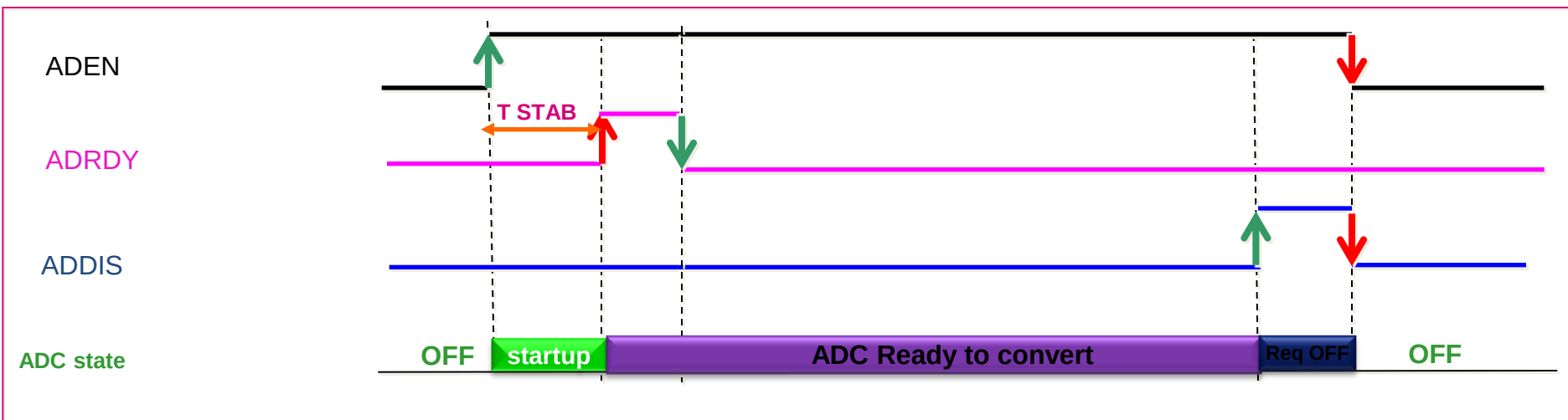


**Note:** The calibration factor is lost when entering Standby, Vbat mode or when the ADC enter deep power down mode. In this case it is possible to re-write the calibration factor into the `ADC_CALFACT` register without recalibrating.

# ADC ON OFF control

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- To enable ADC: Set ADEN=1 then wait till ADRDY flag will be equal to 1,
- What ever is the digital and the analog clock of the ADC, ADRDY signal guarantees that ADC data will be transmitted from one domain to the other.
- ADC cannot be re-programmed unless it is stopped (ADSTART = 0).



# ADC Control bits constraints

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- When ADEN is equal to 0, the software is allowed to write:
  - RCC control bits to configure and enable the ADC clock,
  - The control bits DIFSEL in the ADC\_DIFSEL register,
  - The control bits ADCAL and ADEN in the ADC\_CR register,
- When ADEN is equal to 1 and ADDIS to 0, the software is allowed to write:
  - ADSTART, JADSTART and ADDIS of the ADC\_CR,
  - ADC\_JSQR register
- When ADEN=1 and ADSTART = 0, the software is allowed to write:
  - All control bits related to configuration of regular conversions,
- When ADEN=1 and JADSTART = 0, the software is allowed to write:
  - All control bits related to configuration of injected conversions,
- When ADSTART=JARDSTART=1 and ADDIS=0, The software is allowed to write
  - ADSTP or JADSTP of the ADC\_CR register.

*Note: There is no hardware protection to prevent these forbidden write accesses and ADC behavior may become in an unknown state. To recover from this situation, the ADC must be disabled (clear all ADC\_CR register bits).*

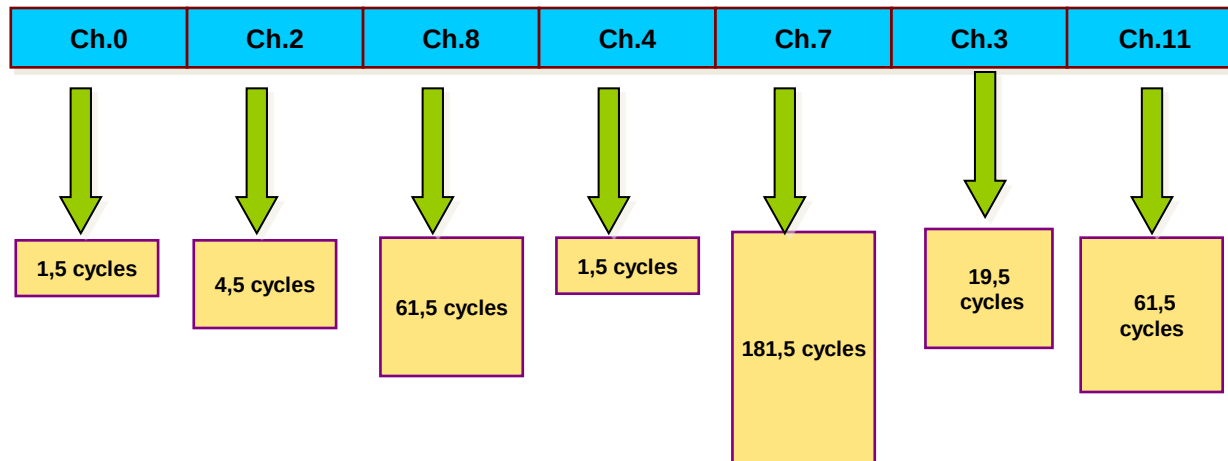
# ADC Channel selection

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- Up to 16 regular and 4 injected conversions with programmable order and programmable sampling time,

**Example:** - Conversion of channels: 0, 2, 8, 4, 7, 3 and 11

- Different sampling time.

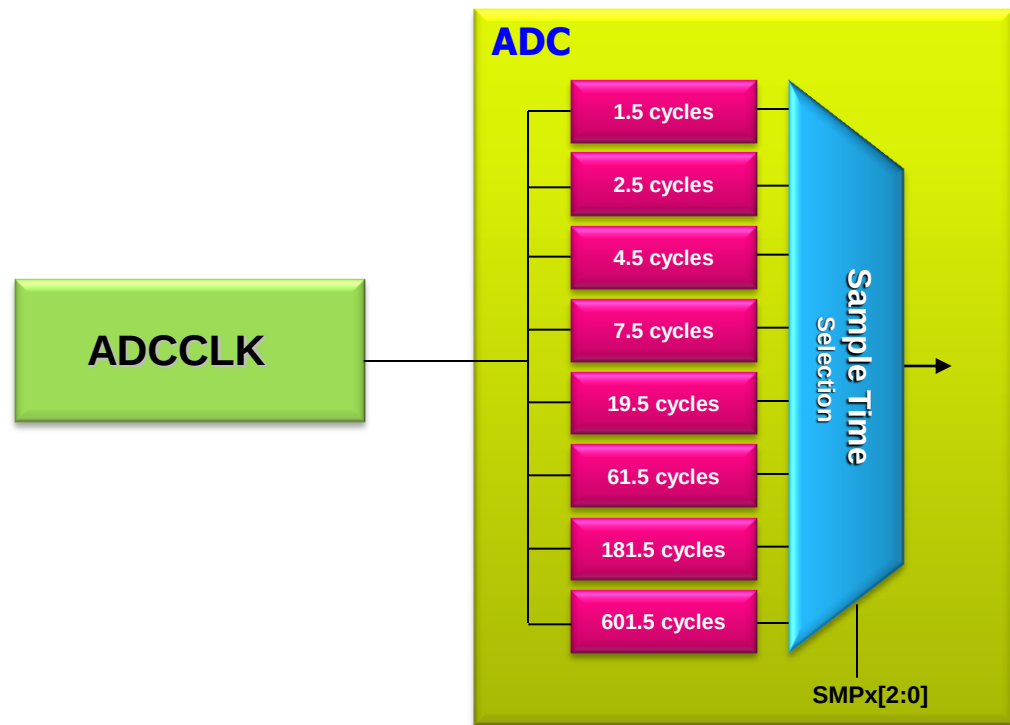


# ADC Sampling Time ( $T_{\text{Sampling}}$ )

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- Three bits programmable sampling time channel by channel programmable:

- 1.5 cycles
- 2.5 cycles
- 4.5 cycles
- 7.5 cycles
- 19.5 cycles
- 61.5 cycles
- 181.5 cycles
- 601.5 cycles



**Note:** The sampling time value depends on the type of channel (fast or slow), the resolution and output impedance of the external signal source to be converted

# Total Conversion Time

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- Total conversion Time =  $T_{\text{Sampling}} + T_{\text{Conversion}}$

| Resolution | $T_{\text{Conversion}}$ |
|------------|-------------------------|
| 12 bits    | 12,5 Cycles             |
| 10 bits    | 10,5 Cycles             |
| 8 bits     | 8,5 Cycles              |
| 6 bits     | 6,5 Cycles              |

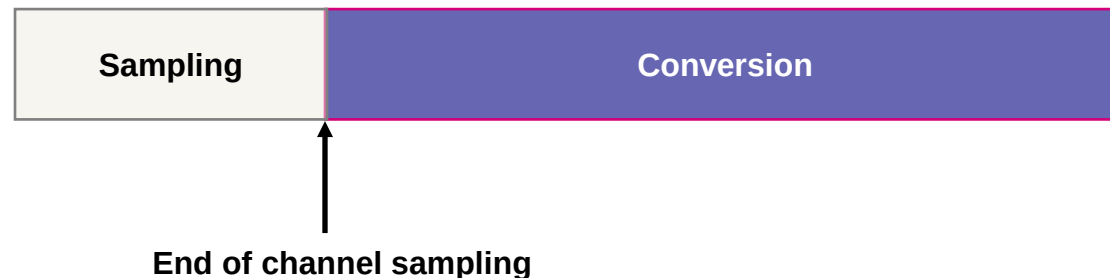
| Resolution | Total conversion Time (When FADC = 72MHz) |                                              |
|------------|-------------------------------------------|----------------------------------------------|
| 12 bits    | $12,5 + 1,5 = 14\text{cycles}$            | $19.4\text{ us} \rightarrow 5,1\text{ Msps}$ |
| 10 bits    | $10,5 + 1,5 = 12\text{ cycles}$           | $16,6\text{ us} \rightarrow 6\text{ Msps}$   |
| 8 bits     | $8,5 + 1,5 = 10\text{ cycles}$            | $13,8\text{ us} \rightarrow 7,2\text{ Msps}$ |
| 6 bits     | $6,5 + 1,5 = 8\text{ cycles}$             | $11,1\text{ us} \rightarrow 9\text{ Msps}$   |



# End of sampling

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- The ADC indicates the end of sampling phase by setting the EOSMP flag only for regular conversion.
- The EOSMP flag is cleared by software by writing 1 to it.
- An interrupt can be generated if the EOSMPIE bit is set in the ADC\_IER register.



➔ As soon as the sampling is completed it is possible to prepare next conversion (for instance switching I/Os) during the conversion phase.

# Single-ended & Differential input channels

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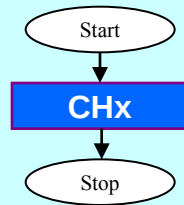
- Channels can be configured to be either single-ended or differential input by writing ADC\_DIFSEL register:
  - In single ended input mode, the analog voltage to be converted for channel “i” is the difference between the external voltage ADC\_INi (positive input) and VREF- (negative input)
  - In differential input mode, the analog voltage to be converted for channel “i” is the difference between the external voltage ADC\_INi (positive input) and ADC\_Ini+1 (negative input)

*Note 1: When configuring the channel “i” in differential input mode, channel “i+1” is no longer usable in single-ended mode or in differential mode and must never be configured to be converted.*

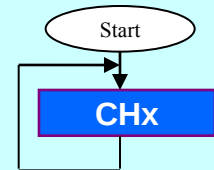
# ADC conversion modes

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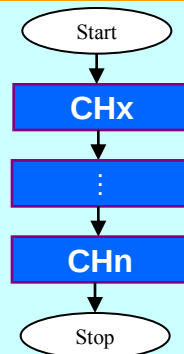
**Single channel  
single conversion mode**



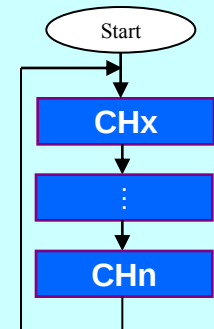
**Single channel  
Continuous conversion mode**



**Multi-channels (Scan)  
single conversion mode**



**Multi-channels (Scan)  
continuous conversion mode**



**Discontinuons conversion  
mode**



# What is the queue of context

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- It is a hardware assistant to prepare the context of the injected channels to allow fast context switching
- A queue of context is implemented to anticipate up to 2 contexts for the next injected sequences of conversions,
- The context consist of:
  - Configuration of the injected triggers (JEXTEN[1:0] and JEXTSEL[3:0]),
  - Definition of the injected sequence (JSQx[4:0] and JL[1:0]),
- Context parameters are defined in ADC\_JSQR register which implements a queue of 2 buffers,

# How to configure the queue of context

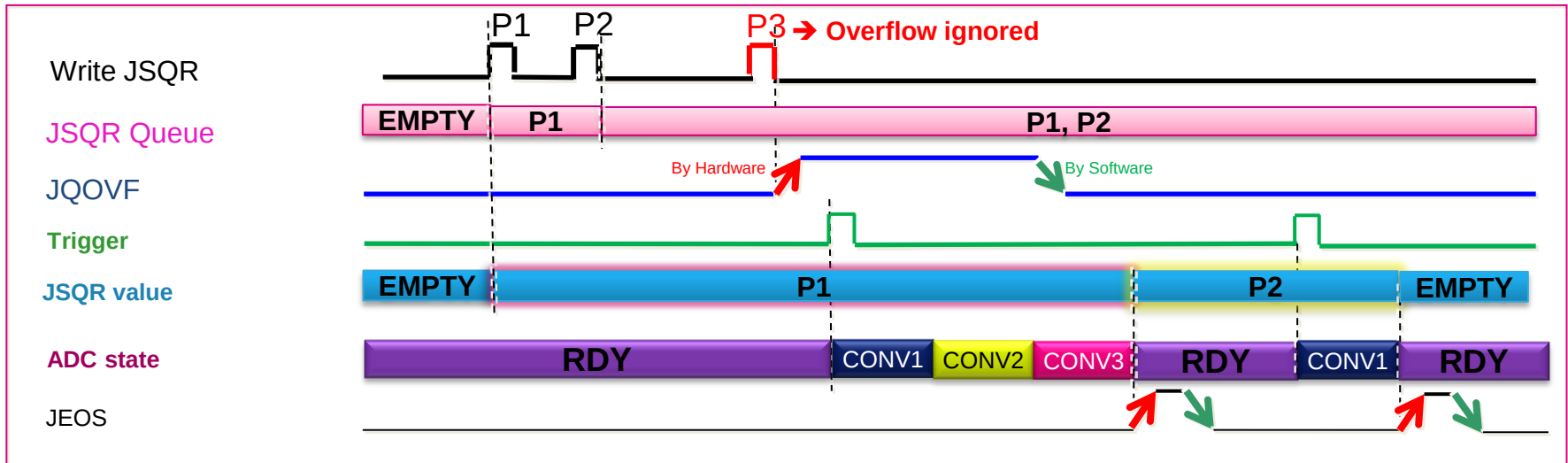
21

- The JSQR register can be written at any moment even when injected conversions are ongoing.
- At the beginning, the Queue is empty and the first write access into the JSQR register immediately changes the context and the ADC is ready to receive injected triggers.
- Once an injected sequence is complete, the Queue is consumed and the context changes according to the next JSQR parameters stored in the Queue.

# Queue overflow

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- A Queue overflow occurs when writing into JSQR register while the Queue is full,
- This overflow is signaled by the assertion of the JQOVF flag,
- When an overflow occurs, the write access of JSQR register which has created the overflow is ignored and the queue of context is unchanged,
- An interrupt can be generated if bit JQOVFIE is set.

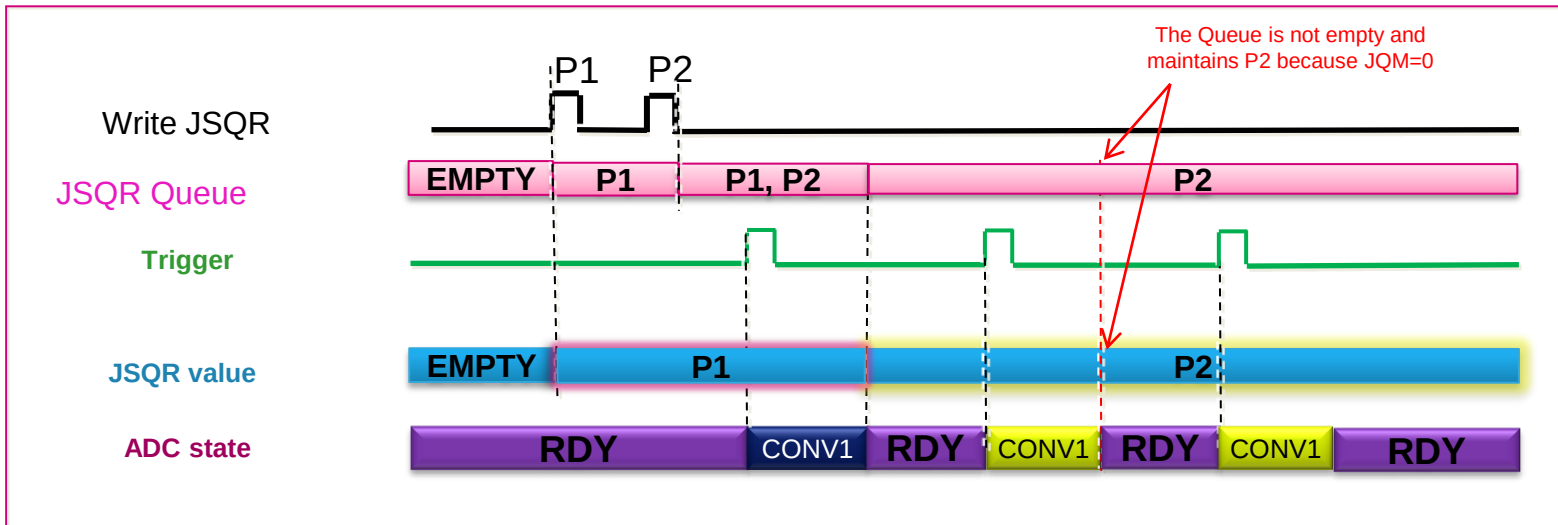


**P1:** sequence of 3 conversions  
**P2:** sequence of 1 conversion  
**P3:** sequence of 2 conversions

# Queue empty, JQM=0

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- When the Queue become empty:
  - If JQM=0 → The Queue is maintained with the last active context,

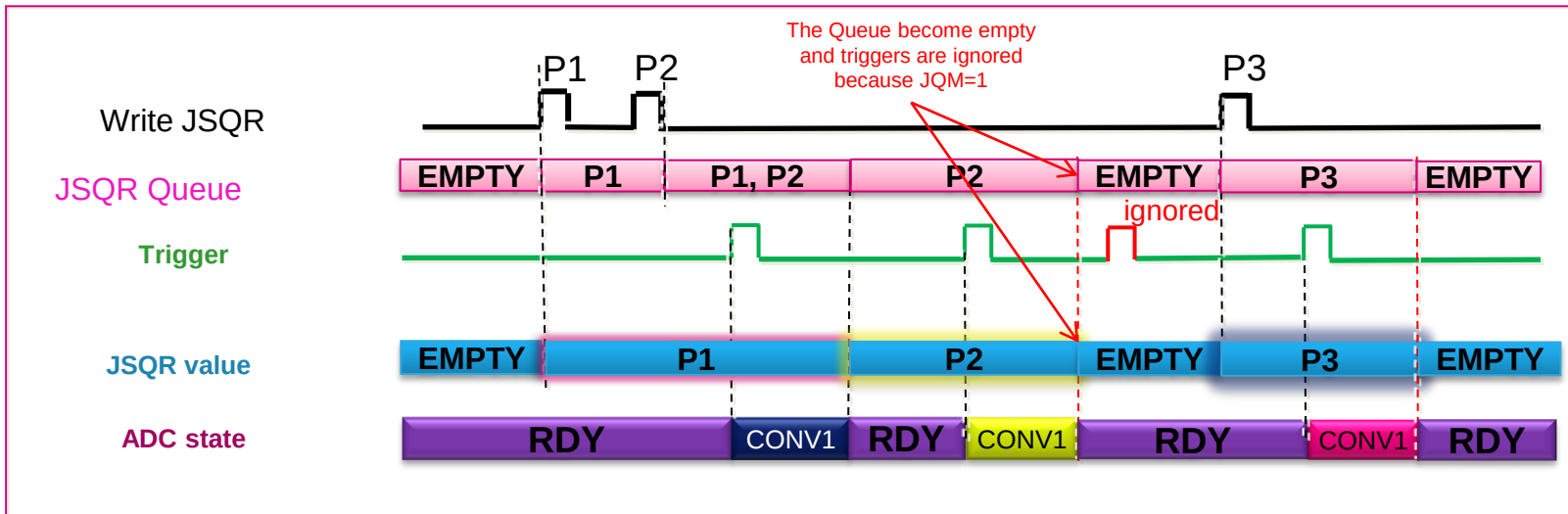


- P1 sequence of 1 conversion
- P2 sequence of 1 conversion

# Queue empty, JQM=1

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- When the Queue become empty:
  - If JQM=1 → The Queue become empty and triggers are ignored,



- P1 sequence of 1 conversion
- P2 sequence of 1 conversion
- P3 sequence of 1 conversion



# ADC Channel offset

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- An offset  $x$  ( $x=1,2,3,4$ ) can be applied to a channel by setting the OFFSET $x$ \_EN of ADC\_OFR $x$  register.
- The channel to which the offset will be applied is programmed into the bits OFFSET $x$ \_CH of ADC\_OFR $x$  register.
- In this case, the converted value is decreased by the user-defined offset written in the OFFSET $x$  bits.
- The result may be a negative value so the read data is signed and the SEXT bit represents the extended sign value.

## Right alignment



Offset disabled, unsigned value



Offset enabled, signed value

## Left alignment



Offset disabled, unsigned value



Offset enabled, signed value

# ADC Overrun management

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- The overrun flag (OVR) indicates a buffer overrun event,
- An interrupt can be generated if the OVRIE bit is set in the ADC\_IER register,
- It is possible to configure if the data is preserved or overwritten when an overrun event occurs by programming the OVRMOD bit:
  - **OVRMOD=0:**

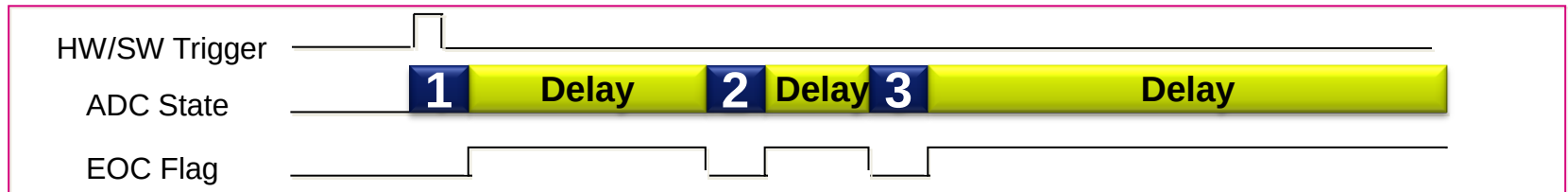
An overrun event preserves the data register from being overwritten: the old data is maintained and the new conversion is discarded. If OVR remains at 1, further conversions can be performed but the resulting data is discarded.
  - **OVRMOD=1:**

The data register is overwritten with the last conversion result and the previous unread data is lost. If OVR remains at 1, further conversions can be performed and the ADC\_DR register always contains the data from the latest conversion.

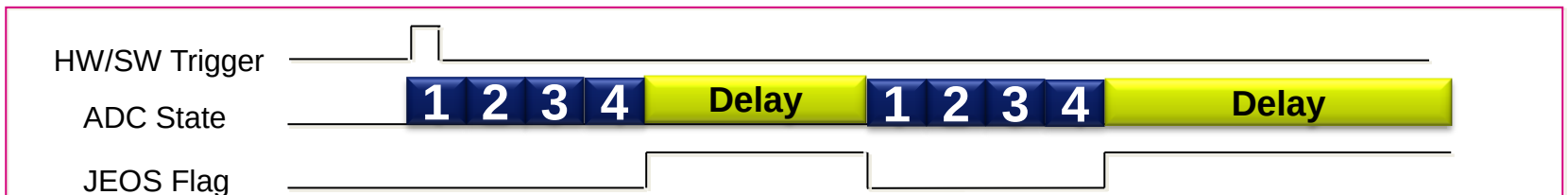
# Auto delayed conversion (1/2)

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- **Auto Delay Mode:** when AUTDLYbits = 1, a new conversion can start only if the previous data has been treated:
  - For regular conversions: once the ADC\_DR register has been read or if the EOC bit has been cleared.



- For injected conversions: when the JEOS bit has been cleared,



Regular channel conversion

**Note :** A trigger event (**for the same group of conversions**) occurring during an already ongoing sequence or during this delay is ignored.

➔ This is a way to automatically adapt the speed of the ADC to the speed of the system that reads the data.

# Auto delayed conversion (2/2)

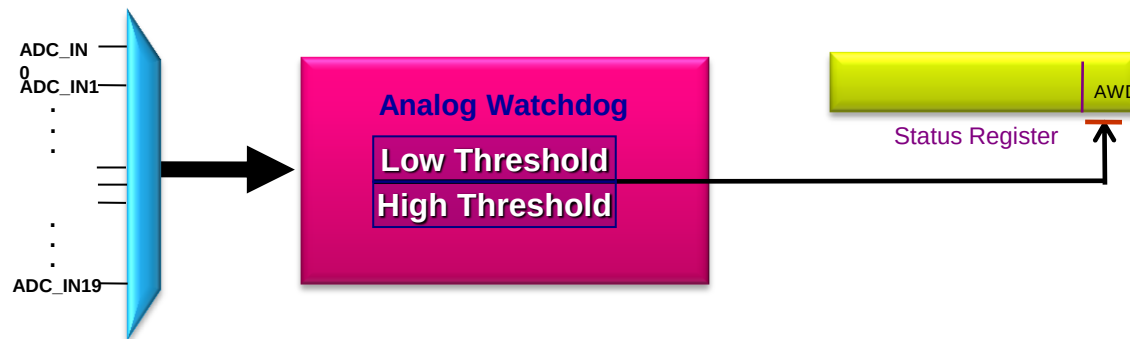
28

- No delay inserted between each conversions of different groups (a regular conversion followed by an injected conversion or conversely)
  - If an injected trigger occur during the automatic delay of a regular conversion, the injected conversion starts immediately,
  - Once the injected sequence is complete, ADC waits the delay of the previous regular conversion before launching new regular conversion,
- In auto-injected mode (JAUTO=1) a new regular conversion can start only when the automatic delay of the previous injected sequence of conversion has ended (when JEOS has been cleared),

# ADC Analog Watchdogs

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- ADC Analog Watchdog 1
  - 12-bit programmable analog watchdog low and high thresholds
  - Enabled on one or all converted channels
  - Interrupt generation on low or high thresholds detection
- ADC Analog Watchdog 2&3
  - Enabled on some selected channels by programming bits in AWDCHx[19:0],
  - Resolution Limited to 8 bits and only the 8 MSBs of the thresholds can be programmed into HTx[7:0] and LTx[7:0]

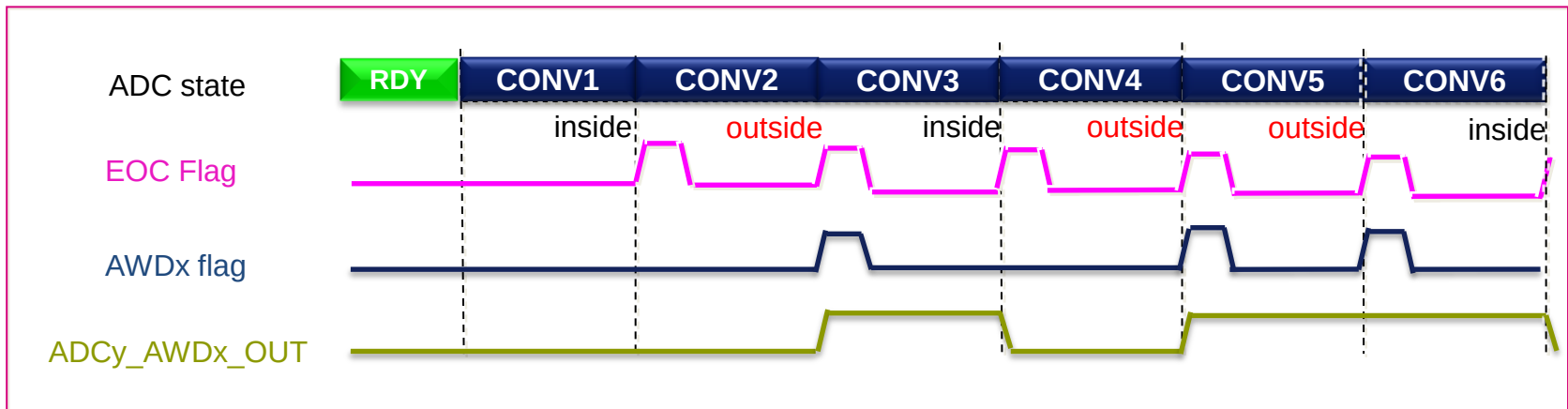


**Note:** The watchdog comparison is performed on the raw converted data before any alignment calculation and before applying any offsets.

# Analog WDG signal generation

30

- Each analog watchdog is associated to an internal hardware signal ADCy\_AWDx\_OUT which is connected to an output timer,



**Note:** AWDx flag has no influence on the generation of ADCy\_AWDx\_OUT (ex: ADCy\_AWDx\_OUT can toggle while AWDx flag remains at 1 if the software did not clear the flag).

# DMACFG bit management

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- DMA can be used to manage the regular channels conversions (ADCx DR register),

- **DMA one shot mode (DMACFG = 0):**

In this mode the ADC stops generating DMA requests once the DMA has reached the last DMA transfer even if conversion has been started again,

- **DMA circular mode (DMACFG=1):**

In this mode, the ADC generates a DMA transfer request each time a new conversion data is available in the data register, even if the DMA has reached the last DMA transfer.

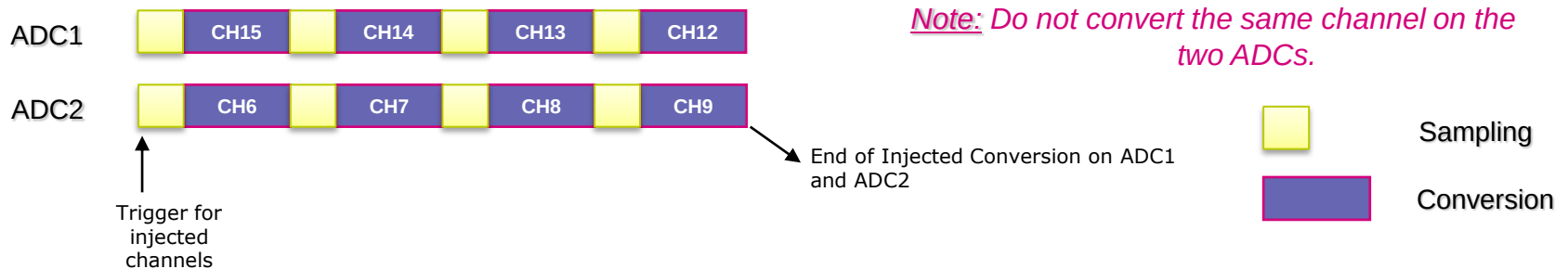
- ADC1 and ADC2 can be used together in Dual mode (ADC1 is the master),
- ADC3 and ADC4 can be used together in Dual mode (ADC3 is the master),
- Six possible modes are implemented:
  - Injected simultaneous mode,
  - Regular simultaneous mode,
  - Interleaved mode,
  - Alternate trigger mode,
  - Injected simultaneous + Regular simultaneous mode,
  - Regular simultaneous + Alternate trigger mode,



# Injected simultaneous mode

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- Converts an injected channel group,
- The external trigger source comes from the injected group multiplexer of the master ADC,
- An JEOC is generated at the end of all channels conversion,
- Results stored on injected data registers of each ADC.

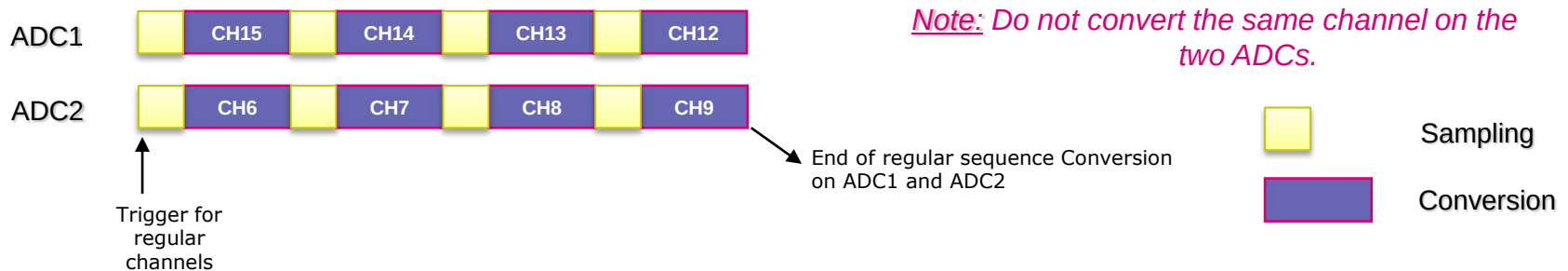


- This mode can be combined with auto-delayed mode,
- Once SW set JADSTART or JADSTP bits of the master ADC, the corresponding bits of the slave ADC are also automatically set,

# Regular simultaneous mode

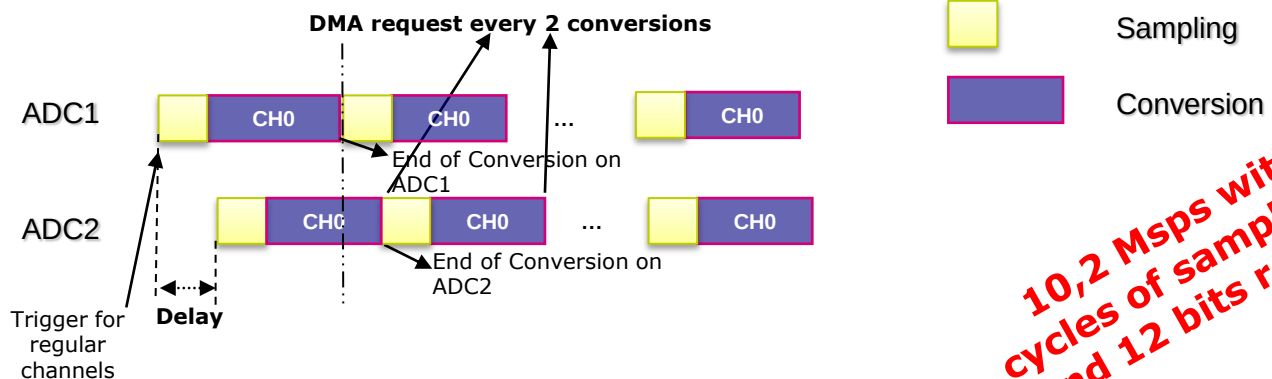
34

- Converts an regular channel group,
- The external trigger source comes from the regular group multiplexer of the master ADC,
- An EOC is generated at the end of each channel conversion,
- Results stored on the common data register `ADC_CDR` and on the each `ADCx_DR`,



- This mode can be combined with auto-delayed mode,
- Once SW set `ADSTART` or `ADSTP` bits of the master ADC, the corresponding bits of the slave ADC are also automatically set,

- Converts a regular channel group (usually one channel).
- The external trigger source, which start the conversion, comes from ADC1:
  - ADC1 starts immediately,
  - ADC2 starts after a configurable delay,
- An EOC is generated at the end of each channel conversion,
- Results stored on the common data register `ADC_CDR` and on the each `ADCx_DR`,



**18 Msps with 1,5  
cycles of sampling time  
and 6 bits resolution**

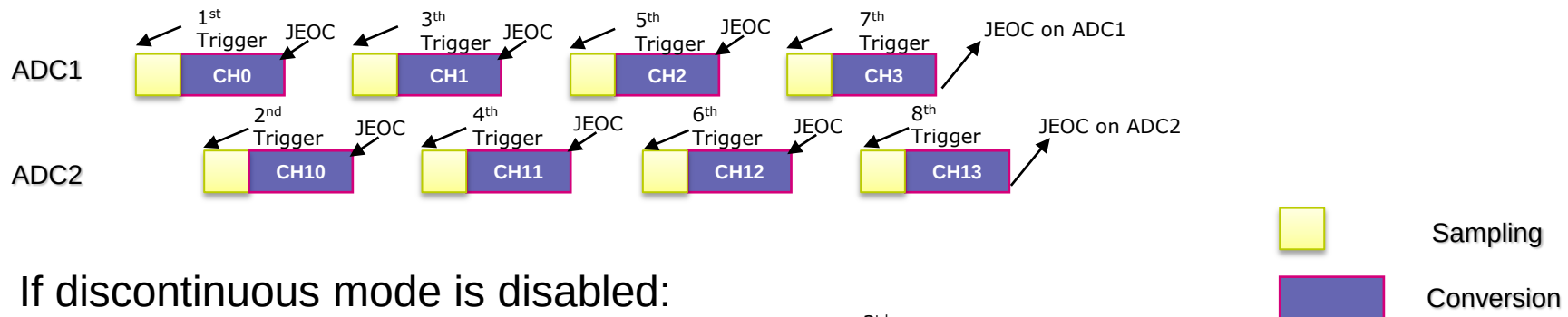
**10,2 Msps with 1,5  
cycles of sampling time  
and 12 bits resolution**

- This mode can not be combined with auto-delayed mode,
- Once SW set `ADSTART` or `ADSTP` bits of the master ADC, the corresponding bits of the slave ADC are also automatically set,

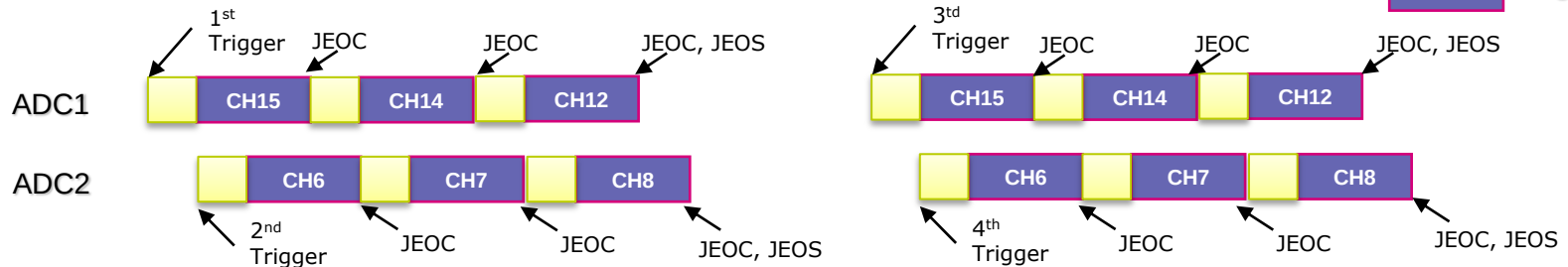
# Alternate trigger mode

36

- Converts an injected channel group,
- The external trigger source comes from the injected group multiplexer of the master ADC,
- If discontinuous mode is enabled:



- If discontinuous mode is disabled:

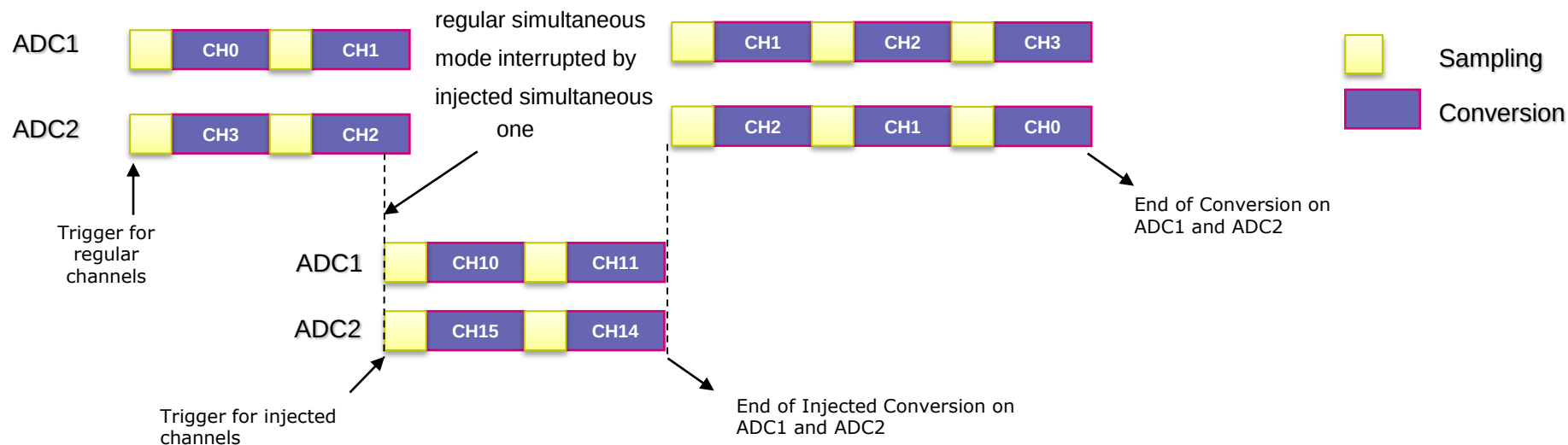


- This mode can not be combined with auto-delayed mode,
- Once SW set JADSTART or JADSTP bits of the master ADC, the corresponding bits of the slave ADC are also automatically set,

# Regular simultaneous + Injected simultaneous

37

- Converts an injected and regular channel groups,
- The external trigger source comes from the master ADC,
- Results of injected channels stored on ADCx\_JDRy registers, and regular channels on each ADCx\_DR register and on ADC\_CDR register.



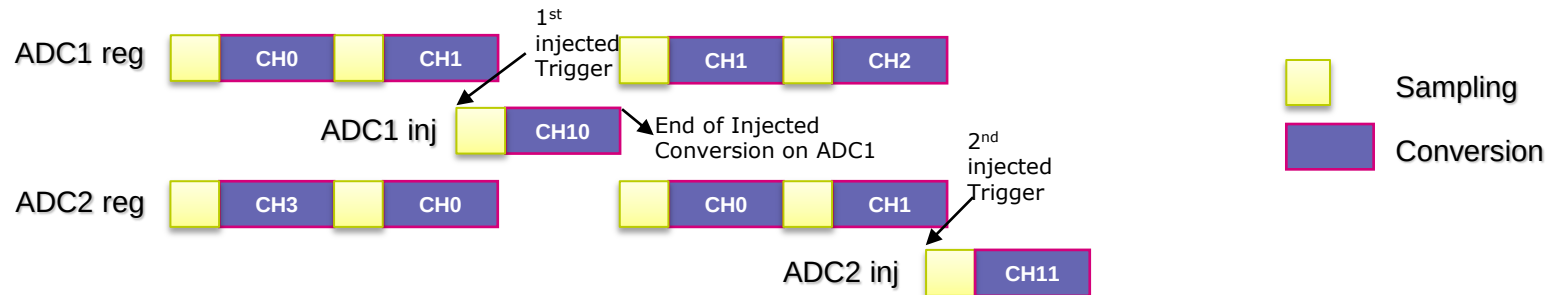
*Note: Do not convert the same channel on the two ADCs.*

- This mode can be combined with auto-delayed mode,

# Regular simultaneous + Alternate trigger

38

- Converts an injected and regular channel groups,
- The external trigger source comes from the master ADC,
- Results of injected channels stored on ADCx\_JDRy registers, and regular channels on each ADCx\_DR register and on ADC\_CDR register.



- This mode can not be combined with auto-delayed mode,

# DMA requests in dual ADC mode

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- **MDMA=0b00:**

- One DMA channel should be configured for each ADC to transfer the data available on ADCx\_DR register,

- **MDMA=0b10:**

- A single DMA request is generated each time both master and slave EOC events have occurred,
  - Used in interleaved and in regular simultaneous mode when ADC resolution is 10 or 12 bits
- 1<sup>st</sup> DMA request    `ADC_CDR[31:0] = SLV_ADC_DR[15:0] | MST_ADC_DR[15:0]`  
2<sup>nd</sup> DMA request    `ADC_CDR[31:0] = SLV_ADC_DR[15:0] | MST_ADC_DR[15:0]`

- **MDMA=0b11:**

- A single DMA request is generated each time both master and slave EOC events have occurred,
  - Used in interleaved and in regular simultaneous mode when ADC resolution is 6 or 8 bits
- 1<sup>st</sup> DMA request    `ADC_CDR[15:0] = SLV_ADC_DR[7:0] | MST_ADC_DR[7:0]`  
2<sup>nd</sup> DMA request    `ADC_CDR[15:0] = SLV_ADC_DR[7:0] | MST_ADC_DR[7:0]`

# ADC Flags and interrupts

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**ADRDY**: « ADC ready »

**EOC** : « **Regular** End Of Conversion »

**EOS** : « **Regular** End Of Sequence »

**JEOC** : « **Injected** End Of Conversion »

**JEOS** : « **Injected** End Of Sequence »

**JQOVF** : «**Injected** Injected context queue overflows»

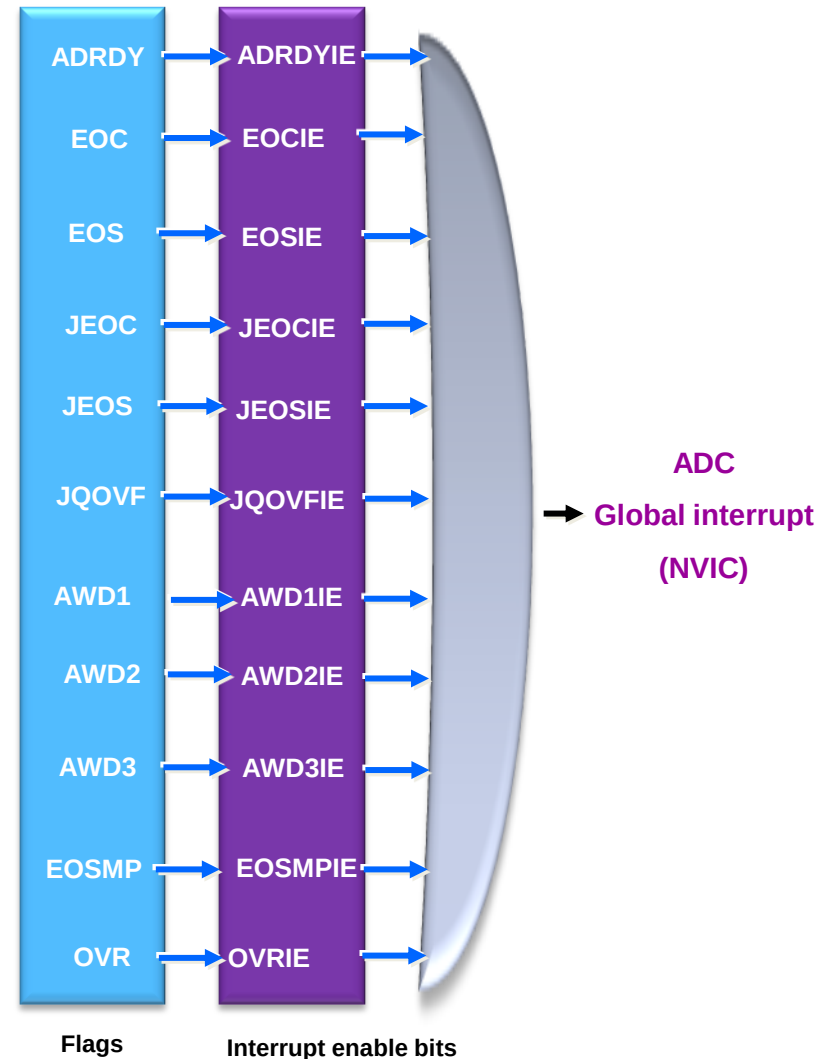
**AWD1** : « Analog watchdog 1»

**AWD2** : « Analog watchdog 2»

**AWD3** : « Analog watchdog 3»

**EOSMP**: End Of Sampling

**OVR**: Overrun





- How many ADC external input channels are in the STM32F3 microcontroller ?

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- What is the max ADC frequency ?

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- What is the queue of context ?

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- How to use DMA in single and in Dual ADC modes ?

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Thank you !

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